Hayden Tam

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Education

York University **June 2028**
Bachelor of Arts Honours in Information Technology *Toronto, ON*

Experience

NRT Technologies **June 2026 – Dec 2026**
QA Engineer Co-op *Toronto, ON*

- Engineered robust **console automation scripts** in **JavaScript** to execute comprehensive regression tests, improving test coverage by **50**%.
- Performed **manual regression, functional, and smoke testing** on mobile and web applications, identifying defects across patron-facing and backend workflows.
- Filed and fixed bug reports in **Jira**, applying severity classification and routing tickets to appropriate teams.
- **Wrote and maintained test plans and test cases** in Qase, organizing test suites for structured regression and functional coverage.

Extern, Outamation **Oct 2025 – Dec 2025**
AI & Outamation Extern *Remote, NY*

- Built an AI-powered pipelines pipelines to analyze over **200+** page mortgage documents across **10+ document types**, leveraging **OCR (Tesseract, PaddleOCR), PyMuPDF, Python, and RAG** for structured **data extraction**.
- Developed an **end-to-end document retrieval system** using **LlamaIndex** and **RAG** for multi-document mortgage blobs, improving retrieval precision by **25%-35%** through chunk tuning, metadata filtering, and **open-source LLM evaluation (Mistral, Phi-2)**. Made technical reports with **deployment** recommendations and a demo UI.

Projects

Memory Hiearchy Simulator | *C++, Python, matplotlib* **Sept 2026**

- Implemented a **3 level cache simulator (L1/L2/DRAM)** modelling data movement and latency during **matrix multiplication** workloads.
- Visualized **bandwidth utilization** per cache level and targets **NPU** style **memory analysis**.

Expression Compiler with LLVM | *C++, LLVM, CMake* **Aug 2026**

- Built a **compiler** for an **arithmetic expression language including lexer, recursive descent parser, AST, and LLVM IR** code generator.
- Produces executable output via **llc** and supports variables and operator precedence.
- Demonstrates **end-to-end compiler pipeline** form source to machine code.

CUDA Matrix Multiplication Benchmark | *C++, CUDA* **Jan 2026**

- Built a **parallel matrix multiplication kernel** using **CUDA**, mapping thread blocks via grid and block indexing.
- Developed **GPU implementations** and **benchmarked** execution time using **CUDA events**.
- Optimized memory access using shared **memory tiling** to reduce global memory latency.

GPU Accelerated Document Q&A System | *Python, PyTorch, FAISS, Transformers, Gradio* **Dec 2025**

- Built a **GPU-accelerated RAG** pipeline to process **100+** scanned and unstructured PDFs **end-to-end**.
- Implemented **semantic search and vector similarity retrieval** using **GPU-accelerated embeddings and FAISS**, reducing query latency by **40%–60%** compared to CPU-only baselines.
- Integrated **Transformer-based language models** for context-aware QA, enforcing **100% citation-grounded responses** to eliminate hallucinations and ensure explainability.

Technical Skills

Languages: C++, Python, Java, SQL, HTML/CSS, JavaScript, MATLAB
Frameworks/Libraries: PyTorch, TensorFlow, CUDA, LLVM, ONNX, CMake, Pandas, NumPy, PySpark, PostreSQL, Scikit-Learn, Flask
Developer Tools: Git, Linux, Bash, Powershell, Jira, Claude, Copilot, Docker, AWS, Kubernetes, GitLab, Qase